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GraalLSP

GraalLSP: A Zero-overhead Polyglot Language Server Implementation for the GraalVM

GraalLSP: Eine Overhead-freie Language Server Implementierung für die GraalVM

by

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Abstract

The english abstract.

Zusammenfassung

Die Zusammenfassung auf deutsch.

Acknowledgments

I owe everything to my cat.

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LSP Language Server Protocol
IDE Integrated Development Environment

1. Introduction

- language development is challenging task
- what makes a good language: amongst performance, std lib, ecosystem and community, tools (for language developers and users) are important
- who provides the tools? language developers vs. language users
- normally tools are developed for a specific language and cannot be reused or need lots of adaptation
- In the Truffle framework, tooling (instrumentation) is part of the framework and therefore tools are implemented language-agnostic once
- language developers and users can use existing tools for free or build new ones and still create high-performance languages
- Truffle has already some langauage-agnostic tools implemented (debugger, cpu-sampler, ...)
- Compared to a full IDE, tools are missing
- Implementing LSP provides further IDE-like features
- Can we implement LSP in Truffle to get IDE for free? With minimal effort for the language implementer and minimal to zero overhead for the language performance?

1.1. Contributions

- We propose GraalLSP - LSP language server for Graal/Truffle
- Aim: integrate LSP in Truffle with:
  - minimal to zero performance overhead
  - minimal API changes to Truffle
  - minimal impl overhead for language developer
  - minimal maintainability overhead for language developer
- Result: Generic Polyglot language server
- -> Language developers get lang-tools (via language server) for free
- -> Language users can choose from wide range of LSP clients
- Contribution in detail:
  - A polyglot language server implementation for Graal/Truffle-implemented languages
  - An exemplary VSCode extension, to use VSCode as language client
  - Proposals to extend the Truffle Framework to provide more language internals needed to support more LSP features

1. *Introduction*

   – An experience report how well existing Graal/Truffle-implemented languages work with GraalLSP

1.2. Structur of this Thesis

2. Context

2.1. Graal and Truffle

2.2. Language Server Protocol

2.2.1. lsp4j - Java binding for LSP

- version (+ LSP version)

2.2.2. LSP client: VSCode example

3. Approach

- language implementaion + tool development = challenging task
- we propose: implement LSP for Truffle

3.1. Which data needs LSP server?

- LSP feature overview
- URIs and Locations
- data needed for: diagnostics, completion, goto-definition, ...

3.2. Use of existing Truffle-APIs

- what is already there?
- parsing sources
- AST
- scopes (global/local)
- frames and frame descriptors
- instrumentation
- -> functionality to support impl of debugger

3.3. What is missing?

3.3.1. Static information

- AST design focused on runtime, lacking many parse-time infos
  - mark declarations
  - mark variable read/write locations
  - ...
  - in general: semanitc relations between nodes
- most lang impls: no syntax-error tolerant parser

3.3.2. Proposed Truffle API changes

- language API
- Tags

3.4. How to deal with dynamically typed languages? -> Using runtime information

- Observe execution via instrumentation
- Side-effects -> Truffle-sandboxing
- Evaluation strategies:
  - Literals
  - Coverage data
    - storing frames
    - source location format
    - migrating coverage data on source edit
  - Run-to-section
  - Globals

3.5. Architecture

- graphic showing launcher, lsp server, truffle api communication
- comparison to other language server impls (built on top of language)

4. Implementation

4.1. The lsp4j library

4.2. Architecture

4.2.1. Truffle-API + Graal-SDK

4.2.2. Launcher, Instrument, Server

4.2.3. Thread with entered Context

4.3. What a TruffleLanguage needs to implement

The onInputValue method can be used in combination with source section filters to intercept very fine-grained execution events such as intermediate values used by language expressions. The intermediate values that are accessible to the instrumentation framework greatly depend on the instrumentation support provided by each language.

4.3.1. Expression- and Statement-Tag

4.3.2. findSourceLocation()

4.3.3. findTopScopes()

4.3.4. findLocalScopes()

4.3.5. findMetaObject()

4.3.6. getCompletionTriggerCharacters()

4.3.7. boxPrimitive()

4.3.8. parseInline()

4.3.9. GetSignature custom message

4.4. LSP features implementation - TextDocumentService

4.5. LSP features implementation - WorkspaceService

4.6. LSP client config

4.6.1. VSCode extension exmaple + Limitations

5. Experience report

5.1. Between IDE and Language Workbench

Language Workbench focus/aim: Efficient language (DSL) development, performance is not main focus vs. Graal/Truffle aim: Efficient execution

Assumption: Language Workbenches use static information only, our Truffle-based solution uses dynamic type information

5.2. Completion response time

5.3. Limitations

own chapter? or in cha:implementation?

5.3.1. VSCode: Need to define supported languages in extension

5.3.2. VSCode: Syntax highlighting defined at client-side

5.3.3. Server: Only TCP (connecting to already running server), no std-in/-out impl

5.3.4. Truffle: Only first syntax error is reported. Lang parsers fail on first error

6. Related Work

6.1. Graal

[5]

6.2. Squimera

[4] [3]

6.3. Language Workbenches

[1]

6.4. Run-time type collection

[2]

7. Conclusion and Future Work

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Appendix A.

First Unimportant stuff.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Eigenständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Arbeit selbständig verfasst sowie keine anderen Quellen und Hilfsmittel als die angegebenen benutzt habe.

Berlin, den 12. Oktober 2018

Daniel Stolpe

Todo list

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